

Risk Score - API Document

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Document Control

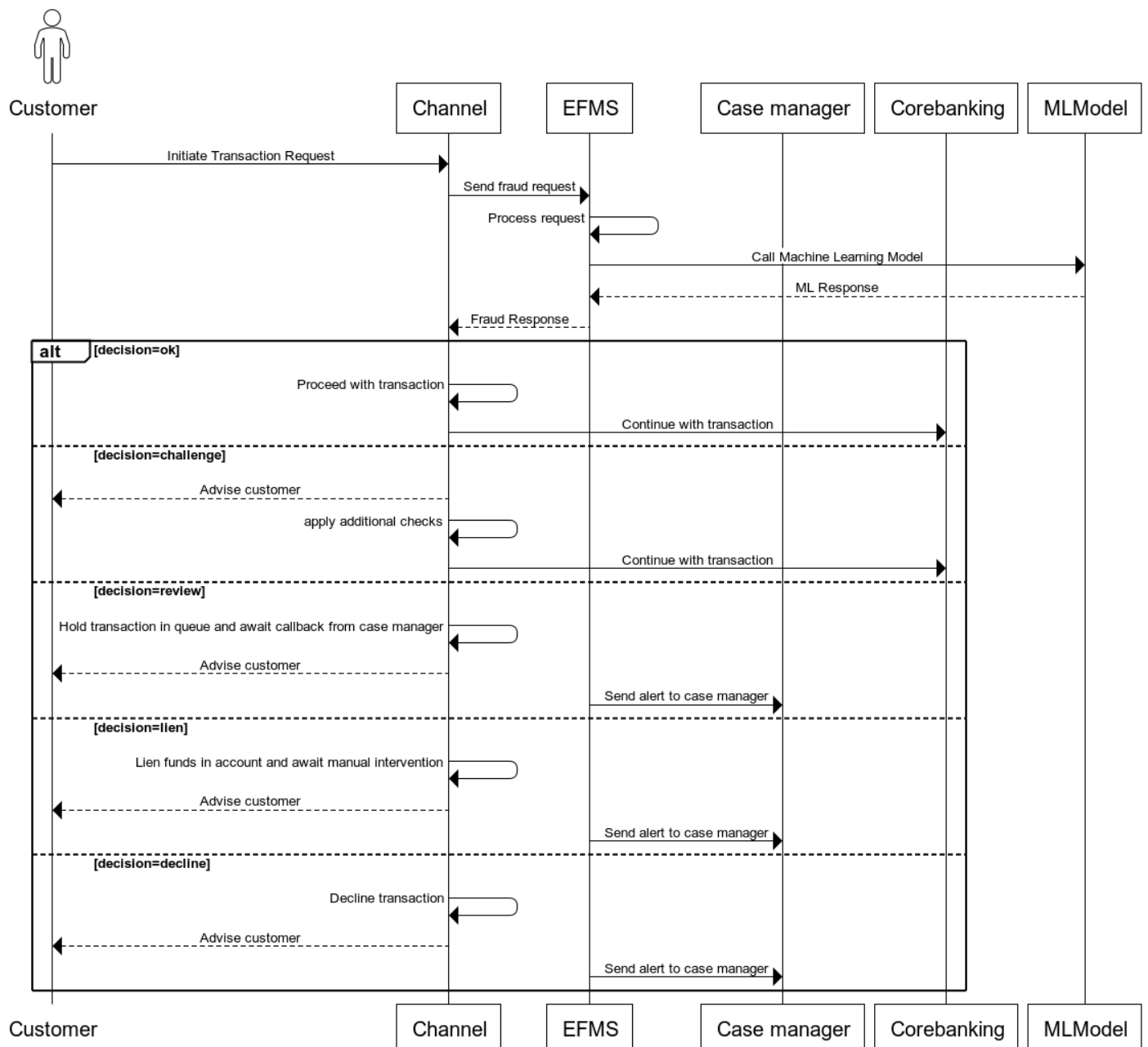
Date	Version	Author	Remarks
26/08/2020	0.0.1	John M	Initial draft

API Description

This API evaluates a transaction against a set of rules and returns a risk score. The originator of the transaction will then use the score to determine if to 1) Decline transaction, 2) Challenge/Advise, or 3) Accept the transaction

Flow

Channel Integration to EFMS



API Summary

API	Fraud Risk score API
Protocol	REST
Format	JSON

Request

Endpoint: /fraud/v1/score
Method: POST

Headers

Header Name	Value
Content-Type	application/json
Authorization	Bearer xxxx

Fields

Field	Description	Type
requestId	Unique request ID	String(50)
customerId	Customer CIFID	String(20)
bankId	Bank ID	INT
transactionType	Type of transaction	String(50)
transactionTime	Date and time of transaction	Date (yy dd'T'HH:MM:ss)
sourceChannel	The origination channel	String(20)
dr.account	Debit account (source of funds)	String(20)
dr.transId	Transaction ID of transaction	String(20)
dr.storeOfValue	Store of value name options [CoreBanking,Card]	String(20)
dr.amount	Amount	Double
dr.currency	Transaction Currency	String(3)
dr.customerName	Name of customer	String(20)
cr.account	Credited account number /wallet	String(20)
cr.transId	Transaction ID of transaction	String(20)
cr.storeOfValue	Credit store of value options [Mpesa,AirtelMoneyKe,Pesalink,RTGS,SWIFT,CoreBanking]	String(20)
cr.amount	Credited amount	Double
cr.currency	Credited Currency	String(3)
cr.transactionDate	Date and time credited	Date
cr.transactionType	Crediting transaction type	String(50)
cr.customerId	Credited customer ID, e.g Mpesa Number	String(50)
cr.customerName	Credited Customer Name	String(50)
cr.channel	Credited channel [Mpesa,AirtelMoneyKe,Pesalink, RTGS,SWIFT,CoreBanking]	String(50)
deviceDetails.ip	IP of originating device	String(20)
deviceDetails.agent	user agent of originating device	String(20)

Field	Description	Type
deviceDetails.location	location coordinates of originating device	String(10
deviceDetails.country	country of the device	String(20

Sample Request

```
{
  "requestId":"2342342",
  "customerId":"5423234234",
  "bankId":"54",
  "transactionType":"SendToWallet",
  "transactionTime":"27-08-2020T12:00:00.0",
  "sourceChannel":"EAZ",
  "dr":{
    "account":"017019334344",
    "tranId":"5456456463",
    "storeOfValue":"CoreBanking",
    "amount":"20000",
    "currency":"KES",
    "transactionDate":"27-08-2020T12:00:00.0",
    "transactionType":"SendToWallet",
    "customerName":"John Doe"
  },
  "cr":{
    "account":"0029323232442",
    "tranId":"5456456463",
    "storeOfValue":"Mpesa",
    "amount":"20000",
    "currency":"KES",
    "transactionDate":"27-08-2020T12:00:00.0",
    "transactionType":"SendToWallet",
    "channel":"Mpesa",
    "customerName":"John Doe"
  },
  "deviceDetails":{
    "agent":"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_13_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/84.0.4147.105 Safari/537.36",
    "ip":"0.0.0.0",
    "location":"",
    "country":"KE"
  }
}
```

Response

Field	Description	Mandatory
status.status	Response status	Y
status.message	Response message	Y
response.customerId	Customer id (passback value)	Y
response.score	Risk score	N
response.matches	Description of matching rules	N
response.decision	Prescribed decision	N
response.status.code	Rule engine status code	N
response.status.message	Rule engine status message	N

Sample Response

```
`{  
  
  "status": {  
    "status": "0",  
    "message": "Success"  
  },  
  "response": {  
    "customerId": "54200343489",  
    "score": 800,  
    "matches": "",  
    "status": {  
      "code": "000",  
      "message": "Success"  
    },  
    "decision": "DECLINE"  
  }  
}  
`
```

Response Codes

Http Status	Status	Message
200	0	Success
500	5	Internal Server Error
400	4	Invalid request
401	3	Invalid Credentials

Decision

Decision	Action
OK	Allow transaction to proceed
CHALLENGE	Apply 2FA or advise customer
REVIEW	Transaction will be reviewed by an analyst and if clean, a callback will send back to the channel (See Callback section below)
DECLINE	Stop transaction

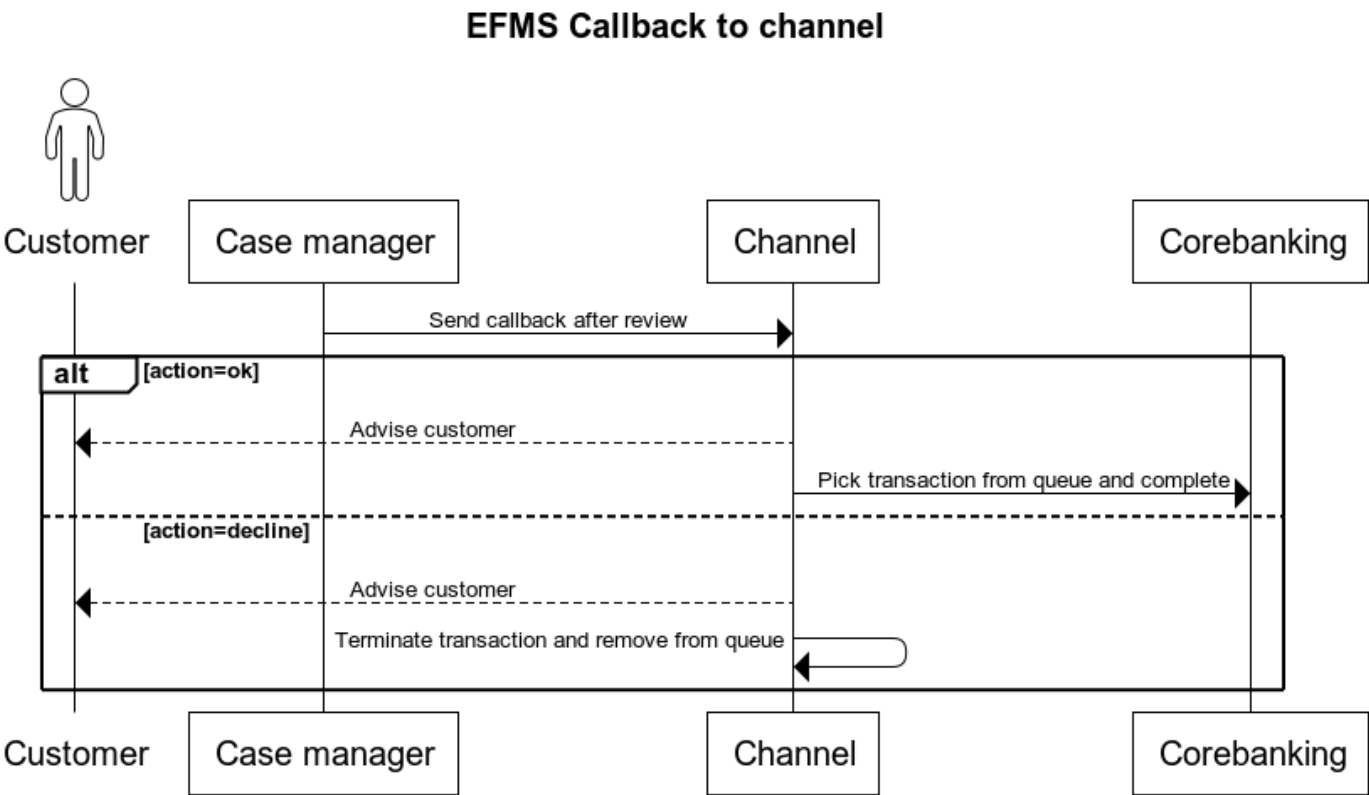
CallBack

For transactions that have the decision of "REVIEW", the channel is supposed to hold the transaction in a queue awaiting the analyst to review the transaction. The outcome of the review may either be:

- 1. OK - Continue with the transaction
- 2. DECLINE - Stop transaction and inform the customer

To enable your channel to receive the callbacks, please implement the following interface

Flow



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Incoming request (Rule Engine --> Channel)

Field	Description	Type	Mandatory
requestId	request Id for the original request	String(50)	Y
action	Prescribed action (see above)	String(20)	Y
comments	Review comments	String(200)	Y

Sample Request

```
`{  
  
  "requestId":"234234343424",  
  "action":"OK",  
  "comments":"customer confirmation recieved"  
  
}`
```

Outgoing Response (Channel --> Rule Engine)

Field	Description	Mandatory
status.status	Response status	Y
status.message	Response message	Y

Sample Response

```
`{  
  
  "status": "0",  
  "message": "Success"  
  
}`
```

Status	Message
0	Success
1	Error - {error message}